

gov51-ps2

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<https://github.com/angelacalderonpio/gov51-ps2.git>

Gov 51 Problem Set 2

1.1 Exploring the Experiment

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(tinytex)
```

Warning: package 'tinytex' was built under R version 4.5.2

```
gay <- read_csv("data/raw/gay.csv")
```

```
Rows: 69592 Columns: 4
-- Column specification -----
Delimiter: ","
chr (1): treatment
dbl (3): study, wave, ssm
```

i Use ``spec()`` to retrieve the full column specification for this data.
i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
##Q1.1
```

```
nrow(gay)
```

```
[1] 69592
```

```
gay |>
  group_by(study) |>
  summarize(n = n())
```

```
# A tibble: 2 x 2
  study      n
  <dbl> <int>
1     1  59207
2     2  10385
```

```
gay|>
  group_by(treatment) |>
  summarize(n = n())
```

```
# A tibble: 5 x 2
  treatment      n
  <chr>          <int>
1 No Contact    37735
2 Recycling Script by Gay Canvasser 6514
3 Recycling Script by Straight Canvasser 6474
4 Same-Sex Marriage Script by Gay Canvasser 12452
5 Same-Sex Marriage Script by Straight Canvasser 6417
```

There are 69,592 total observations in `gay.csv`. The study has 2 unique values (1 or 2). There are 5 treatment conditions, which are: “No Contact,” “Recycling Script by Gay Canvasser,”

“Recycling Script by Straight Canvasser,” “Same-Sex Marriage Script by Gay Canvasser,” or “Same-Sex Marriage Script by Straight Canvasser.”

```
##Q1.2
```

```
gay |>
  filter(study ==1, wave ==1) |>
  group_by(treatment) |>
  summarize(mean_ssm = mean(ssm, na.rm = TRUE))
```

```
# A tibble: 5 x 2
  treatment      mean_ssm
  <chr>          <dbl>
1 No Contact      3.04
2 Recycling Script by Gay Canvasser 3.13
3 Recycling Script by Straight Canvasser 3.01
4 Same-Sex Marriage Script by Gay Canvasser 3.03
5 Same-Sex Marriage Script by Straight Canvasser 3.10
```

The treatment group that shows the highest average support is Recycling Script by Gay Canvasser.

```
##Q1.3
```

```
gay |>
  filter(study ==1, wave ==1) |>
  group_by(treatment) |>
  summarize(sd_ssm = sd(ssm, na.rm = TRUE))
```

```
# A tibble: 5 x 2
  treatment      sd_ssm
  <chr>          <dbl>
1 No Contact      1.68
2 Recycling Script by Gay Canvasser 1.70
3 Recycling Script by Straight Canvasser 1.68
4 Same-Sex Marriage Script by Gay Canvasser 1.67
5 Same-Sex Marriage Script by Straight Canvasser 1.68
```

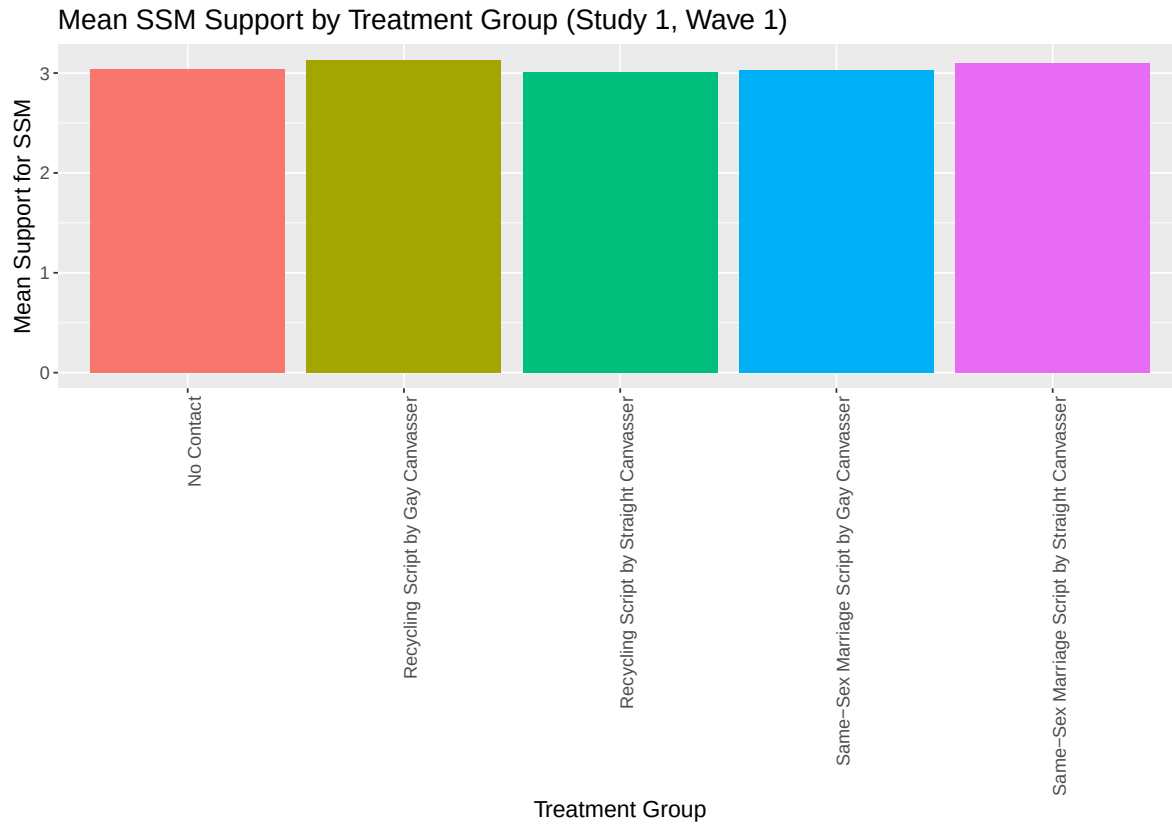
The standard deviations are roughly similar across the five treatment groups, ranging from roughly 1.66 to 1.70. This similarity is important because in a randomized experiment it suggests that the groups have comparable variability in responses as they are being drawn

from the same underlying population and randomly assigned to treatments. If the variability were very different among the groups, it could indicate that the groups were not drawn from comparable populations which would make it more difficult to attribute differences in outcome to the treatment.

```
##Q1.4

bar_plot_gay<- gay |>
  filter(study ==1, wave ==1) |>
  group_by(treatment) |>
  summarize(mean_ssm = mean(ssm, na.rm = TRUE))

ggplot(bar_plot_gay, aes(x= treatment, y= mean_ssm,
                        fill = treatment)) +
  geom_col() +
  labs(x = "Treatment Group", y = "Mean Support for SSM",
       title = "Mean SSM Support by Treatment Group (Study 1, Wave 1)") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  theme(legend.position = "none") +
  theme(text = element_text(size =14))
```



```
##Q1.5
q1_data <- gay |> filter(study ==1, wave ==1)|>
filter(treatment %in% c("No Contact", "Same-Sex Marriage Script by Gay
                        Canvasser"))

q1_data |> group_by(treatment) |>
summarize(mean_ssm = mean(ssm, na.rm = TRUE), sd_ssm = sd(ssm, na.rm = TRUE))
```

```
# A tibble: 1 x 3
  treatment mean_ssm sd_ssm
  <chr>      <dbl> <dbl>
1 No Contact    3.04  1.68
```

```
q1_data
```

```
# A tibble: 5,238 x 4
  study treatment wave  ssm
```

```

      <dbl> <chr>      <dbl> <dbl>
1         1 No Contact      1      5
2         1 No Contact      1      4
3         1 No Contact      1      5
4         1 No Contact      1      4
5         1 No Contact      1      5
6         1 No Contact      1      3
7         1 No Contact      1      2
8         1 No Contact      1      5
9         1 No Contact      1      4
10        1 No Contact      1      1
# i 5,228 more rows

```

```
(3.025195 - 3.042764) / sqrt((1.666099^2 + 1.680540^2)/2)
```

```
[1] -0.01049939
```

The standardized difference is roughly -0.01, which is much smaller than 0.25 in absolute value, suggesting that the “Same-Sex Marriage Script by Gay Canvasser” group and the “No Contact” group are well balanced at baseline. Therefore, the groups appear similar before treatment.

```

##Q1.6
q1_data2 <- gay |> filter(study ==1, wave ==2)|>
filter(treatment %in% c("No Contact", "Same-Sex Marriage Script by Gay
                        Canvasser"))

q1_data2 |>
  group_by(treatment) |>
  summarize(mean_ssm = mean(ssm, na.rm = TRUE))

```

```

# A tibble: 1 x 2
  treatment mean_ssm
  <chr>      <dbl>
1 No Contact    3.04

```

```
(3.139489 - 3.039615)
```

```
[1] 0.099874
```

The difference in means is 0.099874, meaning the gay canvasser group has about 0.10 higher support for same-sex marriage than the no contact group in Wave 2. Because the groups were balanced at baseline, this suggests that this difference at Wave 2 likely reflects the effects of the canvassing treatment (average treatment effect). In other words, this suggests that exposure to the gay canvasser increased support for same-sex marriage relative to no contact.

2.1 The Baseline Distribution

```
##Q2.1
gay_r <- read_csv("data/raw/gayreshaped.csv")
```

```
Rows: 11948 Columns: 6
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): treatment
```

```
dbl (5): study, therm1, therm2, therm3, therm4
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ccap <- read_csv("data/raw/ccap2012.csv")
```

```
New names:
```

```
Rows: 43998 Columns: 3
```

```
-- Column specification
```

```
----- Delimiter: "," dbl
```

```
(3): ...1, caseid, gaytherm
```

```
i Use `spec()` to retrieve the full column specification for this data. i
```

```
Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
* `` -> `...1`
```

```
nrow(gay_r)
```

```
[1] 11948
```

```
ncol(gay_r)
```

```
[1] 6
```

```
colnames(gay_r)
```

```
[1] "study"      "treatment" "therm1"     "therm2"     "therm3"     "therm4"
```



```
nrow(ccap)
```

```
[1] 43998
```

```
ncol(ccap)
```

```
[1] 3
```

```
colnames(ccap)
```

```
[1] "...1"      "caseid"     "gaytherm"
```

For Gay Reshaped, there are 11,948 observations and 6 columns: “study,” “treatment,” “therm1,” “therm2,” “therm3,” and “therm4.”

For Ccap, there are 43,998 observations and 3 columns: “...1,” “caseid,” and “gaytherm.”

```
##Q2.2
```

```
gay_r |>
  filter(study ==1) |>
  summarize(mean = mean(therm1, na.rm= TRUE),
            median = median(therm1, na.rm = TRUE),
            sd = sd(therm1, na.rm = TRUE))
```

```
# A tibble: 1 x 3
  mean median    sd
<dbl> <dbl> <dbl>
1  58.4     52  28.5
```

```
ccap|>
  summarize(mean = mean(gaytherm, na.rm= TRUE),
            median = median(gaytherm, na.rm = TRUE),
            sd = sd(gaytherm, na.rm = TRUE))
```

```
# A tibble: 1 x 3
  mean median    sd
<dbl> <dbl> <dbl>
1  58.7     54  29.4
```

I notice that the summary statistics of the two datasets are very similar. The means, medians, and standard deviations are nearly the same.

#Q2.3

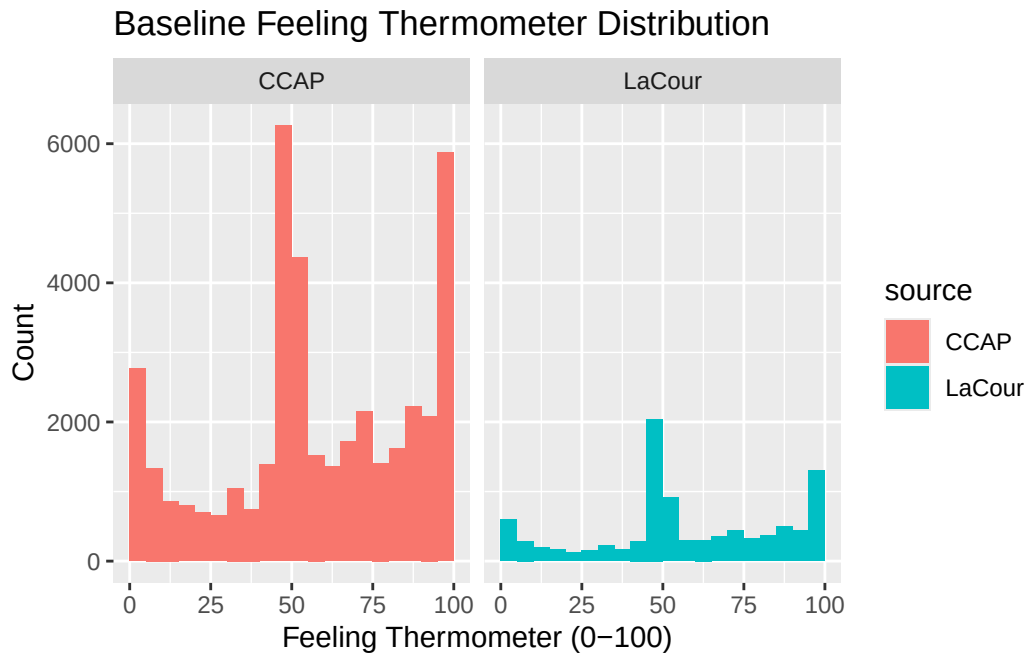
```
lacour <- gay_r |>
  filter(study == 1) |>
  select(therm1) |>
  mutate(source = "LaCour")

ccap_data <- ccap |>
  select(gaytherm) |>
  rename(therm1 = gaytherm) |>
  mutate(source = "CCAP")

both <- bind_rows(lacour, ccap_data)

ggplot(both, aes(x=therm1, fill = source)) +
  geom_histogram(breaks = seq(0,100, by=5)) +
  facet_wrap(~source) +
  labs(title = "Baseline Feeling Thermometer Distribution",
       x = "Feeling Thermometer (0-100)",
       y = "Count")
```

Warning: Removed 3097 rows containing non-finite outside the scale range
(`stat\_bin()`).



The two histograms appear roughly similar in their overall distribution. For example, both show concentrations at the lower and upper ends of the scale and a large share of responses around 50 on the x-axis. Because the shapes look quite similar, it does not appear that the two datasets come from very different populations.

```
#Q2.4
```

```
#Study 1 control group
```

```
study1_control <- gay_r |>
```

```
  filter(study == 1, treatment == "No Contact")
```

```
cor(study1_control$therm1, study1_control$therm2, use = "complete.obs")
```

```
[1] 0.9975817
```

```
sd(study1_control$therm2 - study1_control$therm1, na.rm = TRUE)
```

```
[1] 1.987678
```

```
## Study 2 control group
```

```
study2_control <- gay_r |>  
  filter(study == 2, treatment == "No Contact")  
  
cor(study2_control$therm1, study2_control$therm2, use = "complete.obs")
```

```
[1] 0.9734449
```

```
sd(study2_control$therm2 - study2_control$therm1, na.rm = TRUE)
```

```
[1] 6.605738
```

Study 1: correlation = 0.9976, SD of change = 1.99

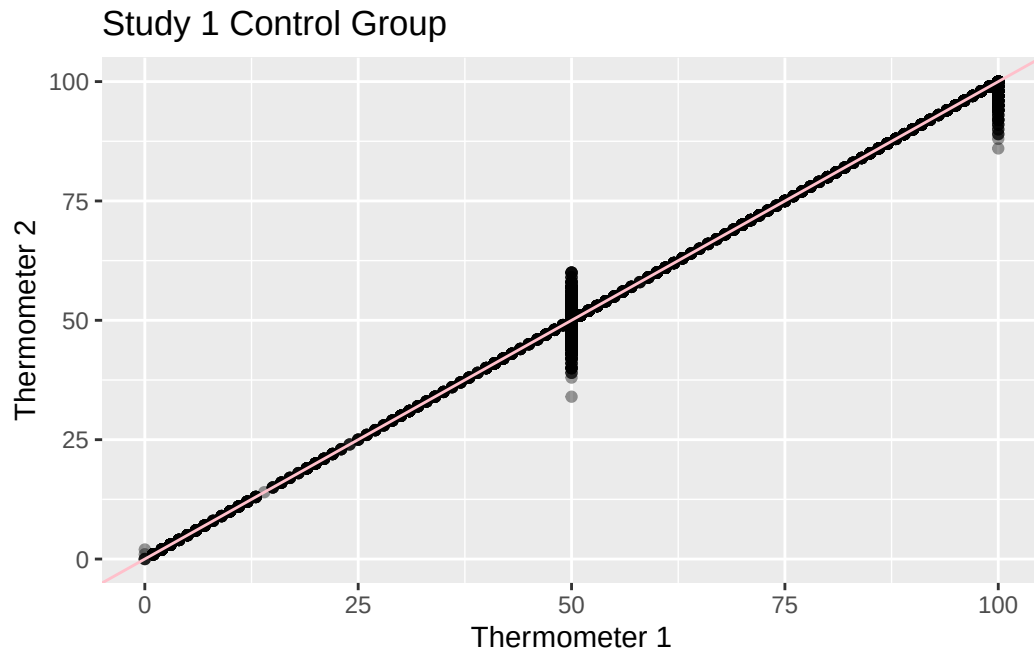
Study 2: correlation = 0.9734, SD of change = 6.61

Study 2 demonstrates a more plausible pattern because its correlation found in this problem (0.973) is within the typical range of 0.85-0.97 for real feeling thermometer data. On the other hand, Study 1's correlation (0.9976) is very close to 1. Furthermore, taking a look at the standard deviation, Study 2's (6.61) is much larger than Study 1's (1.99): this could point to a more realistic variation in responses. Therefore, Study 2 seems more believable.

```
#Q2.5
```

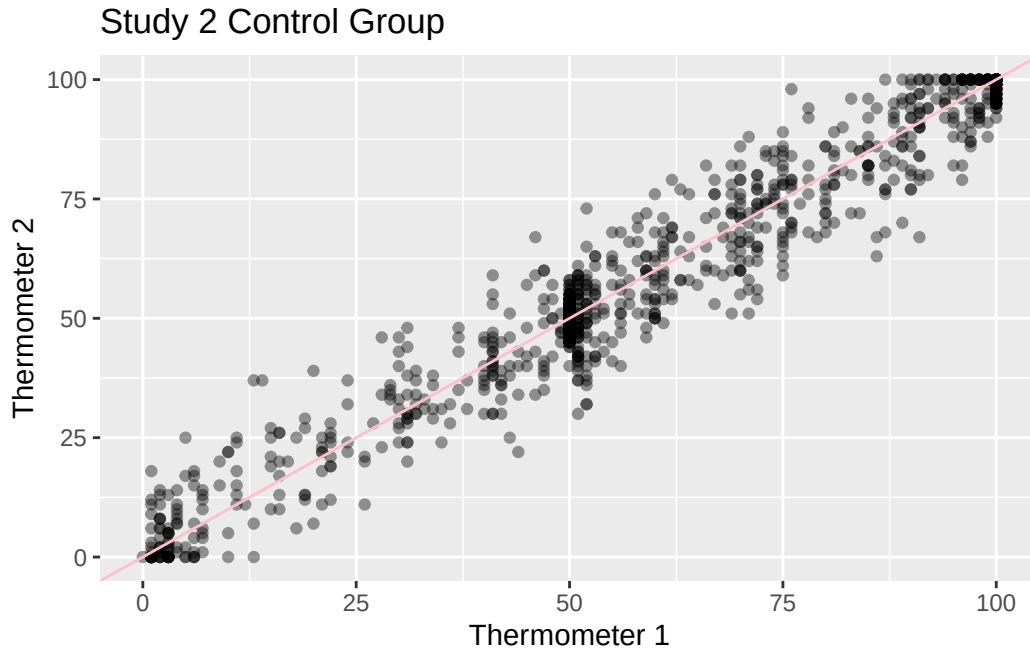
```
ggplot(study1_control, aes(x= therm1, y=therm2)) +  
  geom_point(alpha = 0.4) +  
  geom_abline(slope = 1, intercept = 0, color = "pink") +  
  labs(title = "Study 1 Control Group",  
       x = "Thermometer 1", y = "Thermometer 2")
```

Warning: Removed 568 rows containing missing values or values outside the scale range (``geom_point()``).



```
ggplot(study2_control, aes(x= therm1, y=therm2)) +  
  geom_point(alpha = 0.4) +  
  geom_abline(slope = 1, intercept = 0, color = "pink") +  
  labs(title = "Study 2 Control Group",  
        x = "Thermometer 1", y = "Thermometer 2")
```

Warning: Removed 164 rows containing missing values or values outside the scale range (``geom_point()``).



Study 2 looks more like real longitudinal data. The Study 1 scatter plot depicts the points fitting almost perfectly along the 45 degree line, with the exception of around $x = 0$, 50, and 100, meaning that the responses collected during Wave 2 are nearly the same to those at Wave 1. On the other hand, Study 2's plot demonstrates more spread around the line, meaning that respondents' answers changed somewhat between the two waves. This pattern is more realistic an consistent with actual longitudinal survey data because it's more realistic that responses vary over time.

Q2.6

The findings discovered in question 2 suggest that perhaps the data was fabricated rather than collected in a real experiment. LaCour's findings were suspiciously similar to that of CCAP's national survey. For example, the baseline distributions in Study 1 closely match those of the national survey, which again, should signal some alarm because Lacour claimed to have used his own sample. The shape of the two thermometer histograms are quite similar. Secondly, the test correlation in Study 1 is extremely higher (at roughly 0.998), when typically real feeling thermometer data is around 0.95 to 0.97. And then, the standard deviation of the within-person change was about 1.99, which means that respondents' answers hardly changed in between waves, which isn't realistic because people's mood changes often and can impact their responses. A fabricator might produce data like this because consistent responses realistic-looking baseline distributions can make the results appear convincing.

3.1 Comparing Baselines